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Dynamics of Invariant Network Structure During Seizure Onset and Propagation

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Introduction

- Analysis of seizure transition and propagation is vital for accurate diagnosis and effective treatment of epilepsy. Stereo-electroencephalography (SEEG) records deep brain activity, capturing local neuronal dynamics.

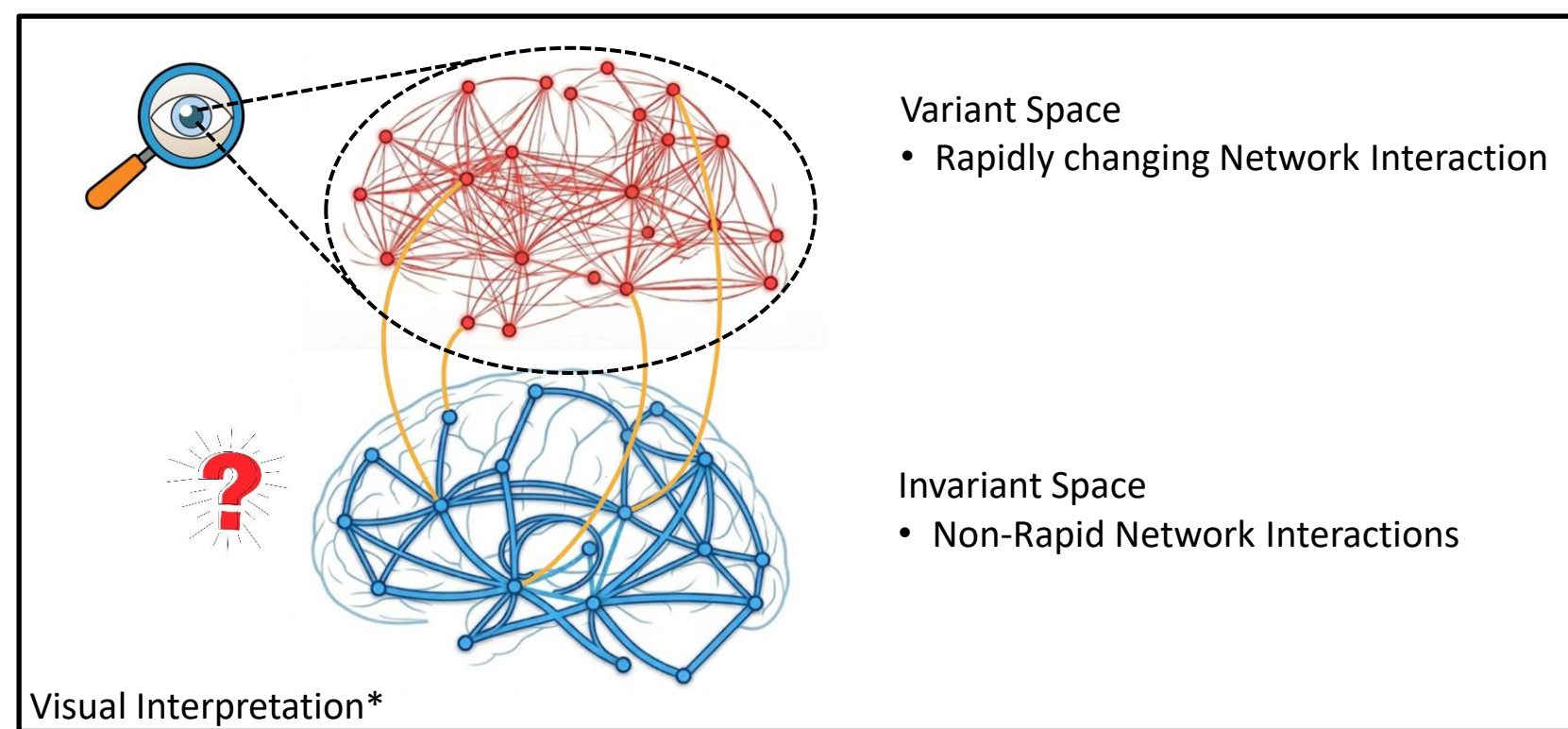


Figure 1: Visual interpretation of the Variant and Invariant Space

- The objective of this study is to develop a pipeline for characterizing the behavior of invariant space and their contribution to the network evolution during seizure propagation.

Methods

Inclusion criterion: Temporal lobe epilepsy (TLE) patients who underwent SEEG implantation between 2020 to 2023. **Seizure free & Non-Seizure free** patients were included.

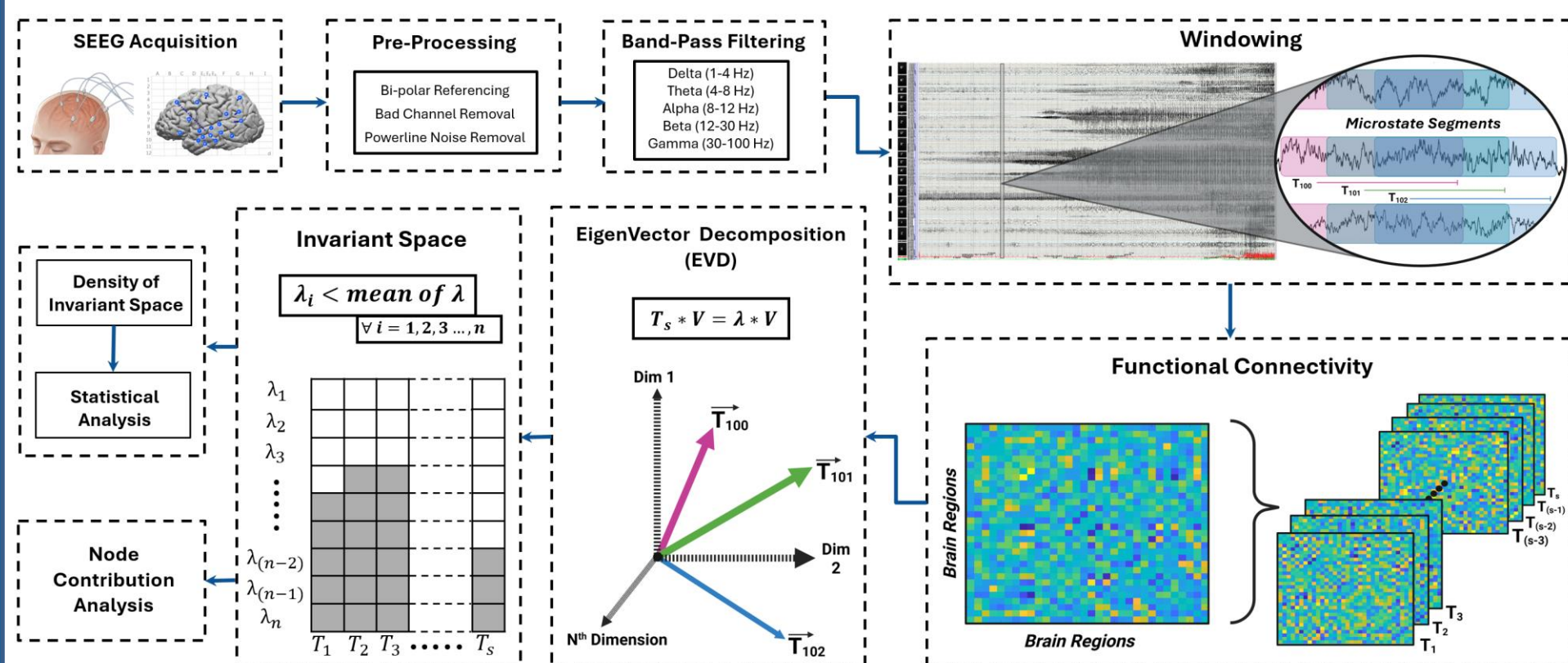


Figure 2: Analysis pipeline demonstrating the flow of methodology. A temporal resolution of 250ms was used in segmenting the SEEG recordings before the estimation of **Phase-Locking Value (PLV)** Functional Connectivity matrices.

- Dataset:** 68 patients (43 Seizure free), 199 seizures (127 from Seizure free group)
- Background:** 2 Background recordings per patient were extracted – **Day & Night**
- Seizure transitions:** Characterized the temporal evolution into 7 states –
 - Preictal** (Includes 120 seconds before the marked onset)
 - Ictal 1-5** (equally divided the Ictal period into 5 segments)
 - Postictal** (Includes 30 seconds after the marked offset)
- Statistical model:** Wilcoxon Paired test was evaluated with FDR correction.

Results

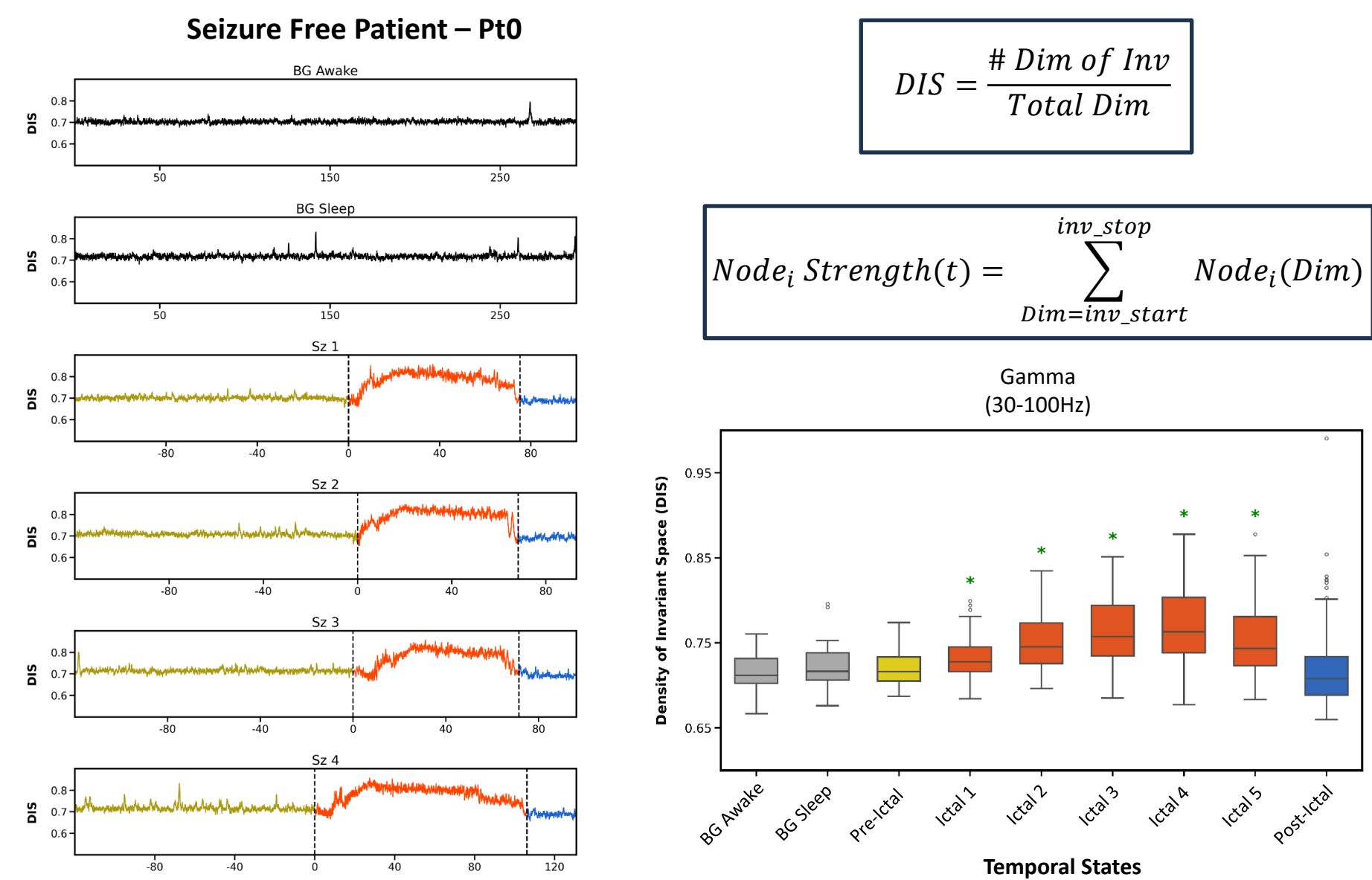


Figure 3: Temporal evolution of Density of Invariant Space (DIS) in the gamma band for a seizure-free patient, showing background activity and four distinct seizures from the same individual.

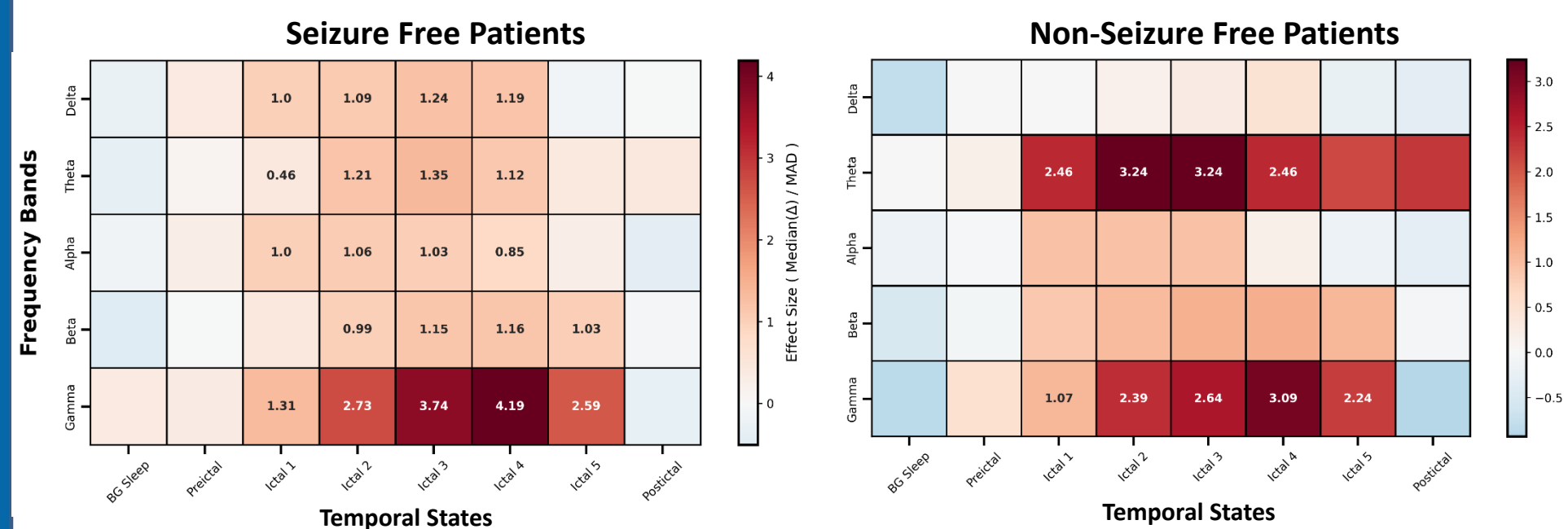


Figure 5: Comparison of DIS values across data segments relative to awake baseline using paired Wilcoxon tests with FDR correction. In seizure-free patients, DIS showed significant effects across all frequency bands during ictal states, whereas non-seizure-free patients exhibited significant differences only in the theta and gamma bands.

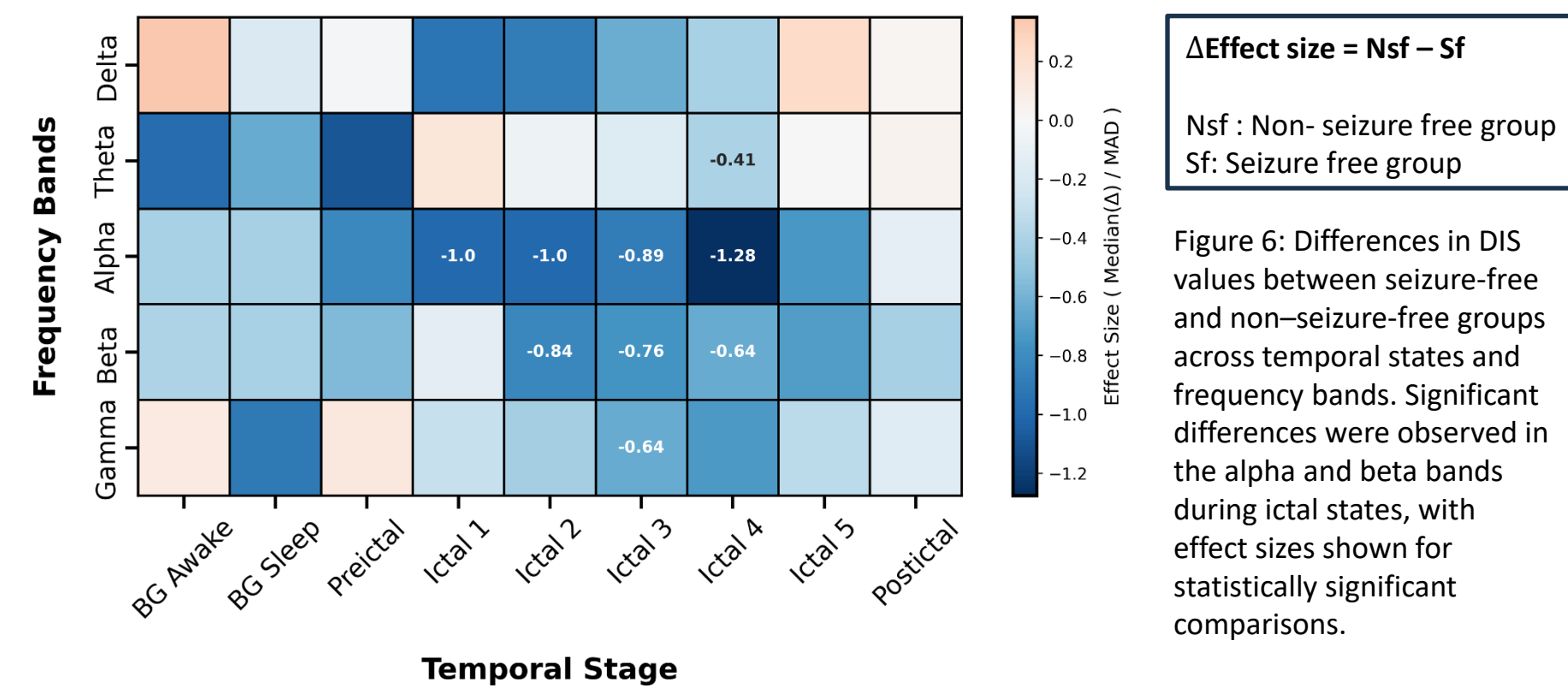


Figure 6: Differences in DIS values between seizure-free and non-seizure-free groups across temporal states and frequency bands. Significant differences were observed in the alpha and beta bands during ictal states, with effect sizes shown for statistically significant comparisons.

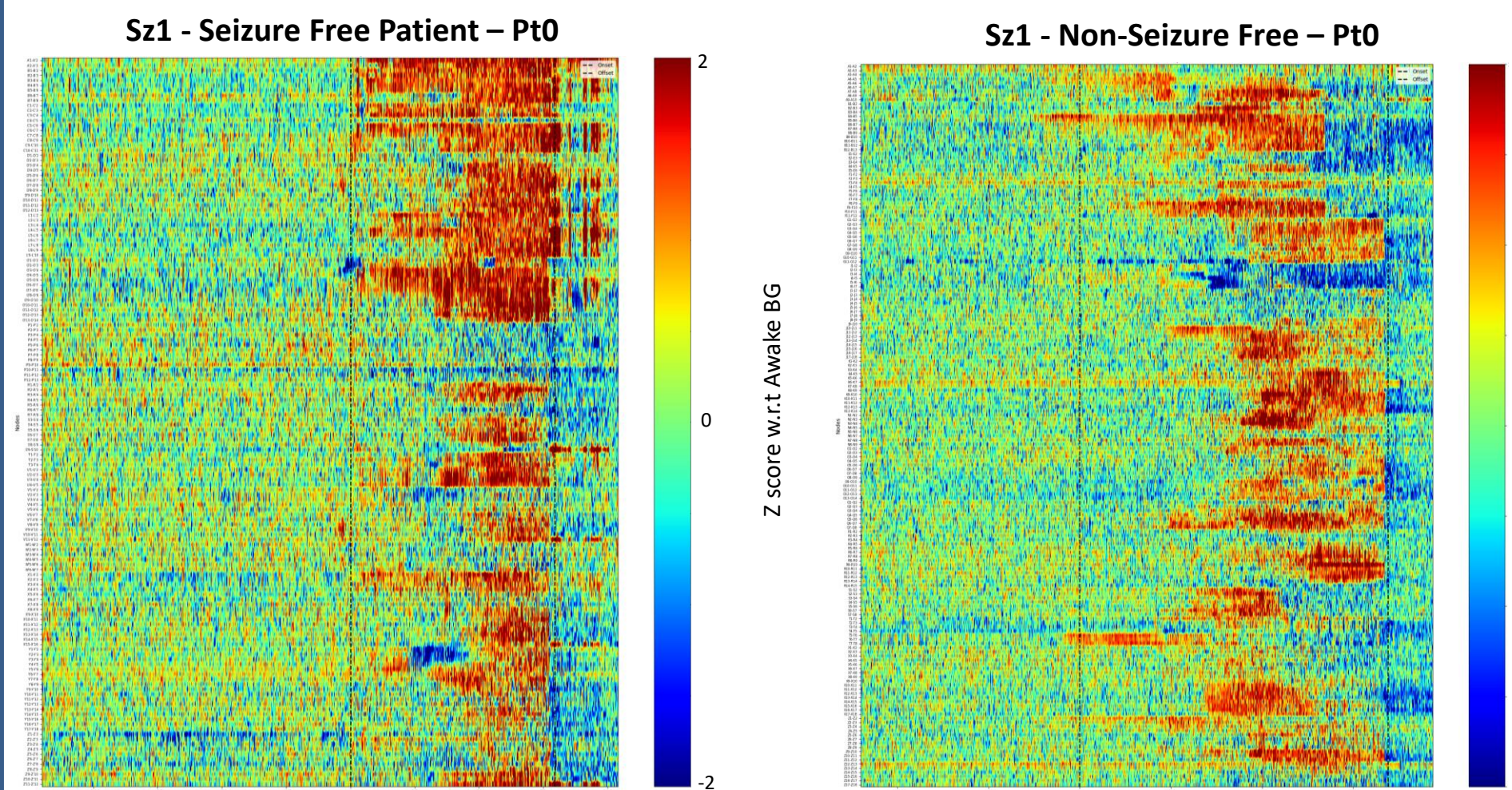


Figure 7: Gamma-band node strength within the invariant space during seizure evolution in both outcome groups. Select nodes show increased strength over time, illustrating the utility of invariant space analysis for characterizing seizure dynamics.

Conclusions

Ictal segments (Ictal 1–5) exhibited a significant elevation in DIS relative to awake baseline ($p < 0.05$), while preictal and postictal periods showed no significant change. This pattern delineates a sharp, state-specific modulation of DIS that accurately isolates seizure onset and termination.

DIS scores demonstrated frequency-dependent divergence between outcome groups across ictal states relative to awake baseline (Fig. 5), as well as in direct intergroup comparisons (Fig. 6). The most robust separations were consistently observed in the alpha and beta bands, indicating these rhythms as key spectral ranges differentiating seizure-free from non-seizure-free ictal dynamics.

Reference:

- [1] Maheshwari et al. (2020). IEEE transactions, vol 28. Pt 8, pp. 1742 – 1749 .
- [2] Dev et al. (2023). IEEE, vol 7, Pt 4, pp. 1149 – 1161.

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